

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Commences proof by considering one side of the identity only: if considering LHS combines terms as a single fraction with a common denominator. If considering RHS writes $\coth \theta$ as a fraction and introduces factor of $1 + \cosh \theta$ to both numerator and denominator Note alternative valid approaches include commencing proof by considering LHS – RHS or LHS divided by RHS	AO2.1	R1	$\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta}$ $= \frac{\sinh^2 \theta + 1 + \cosh^2 \theta + 2 \cosh \theta}{(1 + \cosh \theta) \sinh \theta}$ $= \frac{\cosh^2 \theta + \cosh^2 \theta + 2 \cosh \theta}{(1 + \cosh \theta) \sinh \theta}$ $\because 1 + \sinh^2 \theta \equiv \cosh^2 \theta$ $= \frac{2 \cosh \theta (1 + \cosh \theta)}{(1 + \cosh \theta) \sinh \theta}$ $= \frac{2 \cosh \theta}{\sinh \theta}$
	Explicitly states identity $\cosh^2 \theta - \sinh^2 \theta \equiv 1$ and uses it to eliminate (or introduce) $\sinh^2 \theta$	AO2.4	R1	$\equiv 2 \coth \theta$ AG
	Factorises numerator and cancels correctly for 'their' fraction (if considering RHS rearranges 'their' numerator correctly into two factorised expressions)	AO1.1b	B1F	
	Completes rigorous proof to obtain result AG Only award if they have a completely correct argument, which is clear and contains no slips.	AO2.1	R1	
(b)	Uses result from part (a) to deduce that $\tanh \theta = \frac{1}{2}$	AO2.2a	R1	$2 \coth \theta = 4$ $\tanh \theta = \frac{1}{2}$
	Uses natural log form and substitutes correct value to obtain correct exact form	AO1.1b	A1	$\theta = \tanh^{-1} \frac{1}{2} = \frac{1}{2} \ln 3$
				Total 6 marks

Q2.

- (a) Sketch of $y = \cosh x$

approximately correct with minimum point above the x-axis, symmetrical about y-axis

B1

1

- (b) Attempt to factorise

or complete square or use (correct unsimplified) formula

M1

$$(3\cosh x - 5)(2\cosh x + 1) = 0$$

A1

$$\cosh x \neq -\frac{1}{2}$$

indicated or stated (not merely neglected)

E1

$$x = \ln \left(\frac{5}{3} + \sqrt{\frac{25}{9} - 1} \right)$$

evidence of use of formula. Must see -1 or equivalent

M1

$$= \pm \ln 3$$

ft incorrect factorisation

A1F

A1 for $\pm$

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A1F

6

Alternative:

$$3 \left(\frac{e^x + e^{-x}}{2} \right) = 5$$

$$3e^{2x} - 10e^x + 3 = 0$$

(M1)

$$(3e^x - 1)(e^x - 3) = 0$$

Correct factors

(A1F)

$$x = \ln \frac{1}{3} \text{ or } \ln 3$$

for both

(A1F)

NB if $\cosh x = \frac{e^x + e^{-x}}{2}$ used initially, M0 until quartic in e^x is factorised

M1 for $e^x - 3$ is a factor A1 if correct

M1 for $3e^x - 1$ is a factor A1 if correct

A1 for $x = \pm \ln 3$

E1 for showing remaining quadratic has no real roots

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Q3.

(a)
$$\frac{(e^x + e^{-x})}{2} \frac{(e^y + e^{-y})}{2} - \frac{(e^x - e^{-x})}{2} \frac{(e^y - e^{-y})}{2}$$

M0 if sinh and cosh confused

M1 for formula quoted correctly

M1A1

Correct expansions

Use of e^{xy} A0

A1

$$= \frac{1}{2} (e^{x-y} + e^{-(x-y)}) = \cosh(x-y)$$

AG

A1

4

$$\cosh(x - \ln 2) = \cosh x \cosh(\ln 2) - \sinh x \sinh(\ln 2)$$

(b) (i)

M1

$$\left. \begin{aligned} \cosh(\ln 2) &= \frac{5}{4} \\ \sinh(\ln 2) &= \frac{3}{4} \end{aligned} \right\} \text{any method}$$

Both correct

B1

$$\frac{5}{4} \cosh x = \frac{7}{4} \sinh x$$

A1F

$$\tanh x = \frac{5}{7}$$

AG

A1

4

Alternative:

$$\frac{e^{x-\ln 2} + e^{-x+\ln 2}}{2} = \frac{e^x - e^{-x}}{2} \quad M1$$

$$e^{x-\ln 2} = \frac{e^x}{2} \quad \text{Or} \quad e^{-x+\ln 2} = 2e^{-x} \quad e^{-x+\ln 2} = 2e^{-1}$$

used B1

$$e^x = \sqrt{6} \quad A1$$

$$\tanh x = \frac{5}{7} \quad A1$$

(ii)

$$x = \frac{1}{2} \ln \left(\frac{1 + \frac{5}{7}}{1 - \frac{5}{7}} \right) \quad \text{or} \quad \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{5}{7}$$

Could be embedded in (b)(i)

M1

$$= \frac{1}{2} \ln 6$$

A1

2

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Q4.

(a) $\text{LHS} = \frac{1}{4}(e^x + e^{-x})^2 - \frac{1}{4}(e^x - e^{-x})^2$

M1

Correct expansion of either square

A1

Shown equal to 1

AG

A1

3

(b) (i) $8 \cosh^2 x - 3$

B1

1

- (ii) Sketch of $y = \cosh x$
Must cross y-axis above x-axis

B1

1

- (iii) $\cosh x = (\pm)1.25$
OE; ft errors in (b) (i); allow $\pm$ missing

B1F

$$x = \ln \left(1.25 + \sqrt{1.25^2 - 1} \right)$$

M1

$$= \ln 2$$

A1F

$$\ln \frac{1}{2} \text{ by symmetry}$$

Accept – $\ln 2$ written straight down

Alternatively, if solved by using

$e^{2x} - 2.5e^x + 1 = 0$, allow M1 for

$$x = \ln \left(\frac{2.5 \pm \sqrt{2.5^2 - 4}}{2} \right)$$

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A1F

4

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Q5.

(a) $\frac{9(e^x - e^{-x})}{2} - \frac{e^x + e^{-x}}{2}$

M0 if $\cosh x$ mixed up with $\sinh x$

M1

$$= 4e^x - 5e^{-x}$$

AG

A1

2

- (b) Attempt to multiply by e^x

M1

$$4e^{2x} - 8e^x - 5 = 0$$

A1

$$(2e^x - 5)(2e^x + 1) = 0$$

*ft provided quadratic factorises
(or use of formula)*

M1

$$e^x \neq -\frac{1}{2}$$

PI but not ignored

E1F

$$e^x = \frac{5}{2}$$

A1F

$$\tanh x = \frac{\frac{5}{2} - \frac{2}{5}}{\frac{5}{2} + \frac{2}{5}} = \frac{21}{29}$$

$$x = \frac{\sinh x}{\cosh x}$$

*M1 PI for attempt to use
or equivalent fraction*

M1A1F

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Q6.

- (a) Use of $\cosh^2 x = 1 + \sinh^2 x$

Must be correct for M1

M1

$$4 \sinh^2 x - 7 \sinh x + 3 = 0$$

A1

$$(4 \sinh x - 3)(\sinh x - 1) = 0$$

Provided quadratic factorizes

A1 ✓

$$\sinh x = \frac{3}{4} \text{ or } 1$$

A1 ✓

4

(b) Use of formula for $\sinh^{-1}$

M1

$$x = \ln 2 \text{ or } \ln(1 + \sqrt{2})$$

A1 ✓

A1 ✓

3

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Examiner reports

Q2.

The sketch of $y = \cosh x$ was almost invariably answered correctly in part (a). The most common approach to part (b) was to solve the quadratic equation and then to either use the formula for $\cosh^{-1} x$ in its logarithmic form in which case the solution $-\ln 3$ was generally lost, in spite of the hint given in part (a), or to express the equation $\cosh x = 5/3$ in terms of e^x in which case both answers were usually correctly obtained. Some students chose to express the quadratic equation in terms of e^x , but this led to a quartic in e^x , which they were then unable to solve. The reasons for rejecting the solution $\cosh x = -1/2$ were generally adequate.

Q3.

The bookwork in part (a) was generally well done, and errors if they did occur were almost always when candidates thought that, for instance, $e^x e^y$ was e^{xy} . In part (b)(i), those candidates who took the hint given in part (a) were more successful than those who converted the given equation into exponential form, the latter being unable to handle expressions such as $e^{x - \ln 2}$. Part (b) was usually correct.

Q4.

Part (a) was generally well done apart from a few candidates who were unable to square $\frac{1}{2}(e^x - e^{-x})$ successfully.

Parts (b)(i) and (b)(ii) likewise were well done although $\pm \frac{1}{2}\pi$ did appear on the x -axis of some sketches in part (b)(i). However in part (b)(iii), those candidates using the logarithmic formula for $\cosh^{-1} x$ from the formulae booklet arrived at a single value of x , namely $\ln 2$, but failed to realise that the sketch in part (b)(ii) was intended to give them a hint that $-\ln 2$ was also a solution of the equation. On the other hand, candidates who used the exponential form of either $\cosh x$ or $\cosh^2 x$ automatically produced both answers provided their working was correct, but some of these candidates were unable to handle the algebra leading to a quadratic equation in either e^x or e^{2x} .

Q5.

Almost all candidates scored the two available marks in part (a). However in part (b) a number of candidates did not draw on the hint of part (a) but instead tried to manipulate the equation given in part (b). A few of these candidates expressed the given equation in $\tanh x$ and $\operatorname{sech} x$ and then squared, obtaining a quadratic in $\tanh x$. When factorised these candidates obtained two values for $\tanh x$, only one of which was the correct one. However, virtually no one rejected the incorrect solution so that it was almost impossible to award full marks when this method was used.

Q6.

Apart from a few candidates who factorised the quadratic in $\sinh x$ incorrectly, most candidates worked part (a) correctly. However, many candidates spent more time on part (b) than was necessary. They expressed $\sinh x$ in exponential form and solved the

ensuing quadratic equations rather than quote the formula for $\sinh^{-1}x$ given in the formulae booklet which they were entitled to do. This method also led to superfluous incorrect solutions which candidates needed to reject.



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